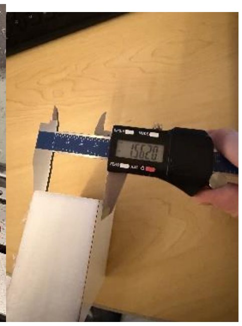
Customer: Keene State	Student Practice Block	April 2026	
Note: This SOP is designed for part production of the student practice block using the Haas Mini Mill CNC machine	Part/Dwg. No.01	Jesse Santoro	

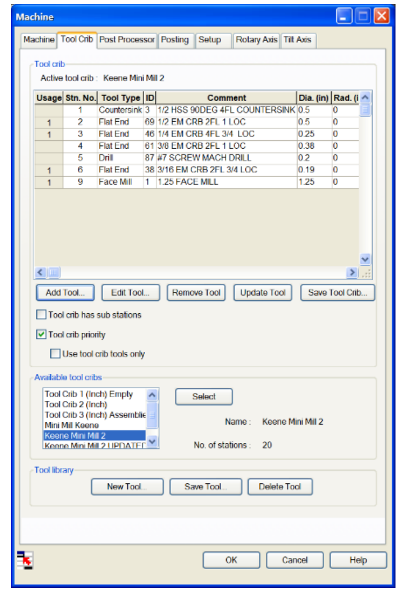
Setting up Haas Mini-Mill
Part No. 01

Setting up tools and machines tool table

Vice Setup-



Tool Crib Setup-



#1 Power Up Machine & Run Spindle warmup program (Should be near the top of the list the file is from 2020)
#2 Once spindle warmup operations are completed open vice *Use the vice key located on the table to the left of the machine
#3 Locate two parallels on the table next to the machine place them into the vice one on either side
#4 Place stock on top of parallels plates use vernier calipers to measure from Z = 0 of your stock checking to make sure no tools will collide with the Vice
#5 Tighten vice until your stock can no longer move from side to side your parallels should still be able to move freely when pushed by hand

Zeroing Part-

#1 Press the hand jog key and move the tool up in the positive Z axis until you are clear of the part
#2 Press the MDI key and enter the command T8 (T+8+ATC Forward) This will switch to the edge finder tool
#3 Set the RPM of your edge finder (Max 750 door open) if you plan to operate with the door closed you can go up to 1000
#4 *OPTIONAL You can trick it to run the edge finder with the door open if you hold down the “CW” key during operation I suggest doing this to improve visibility
#5 Use the edge finder tool to locate the X and Y faces of the part flick the edge finder while in motion to create a slight wobble then using the Hand Jog mode and jog wheel to bring it up to the edge of the part on the desired face
#6 Once you have the desired face located switch back to MDI mode and select the offset button this should highlight the work offset box
#7 Use the arrow keys to select the desired axis in the menu and Press Key “Part Zero Set”
#8 Switch back to hand jog mode increase Z so your edge finder sits above the part then move your edge finder over the part until it reads 0.01 from your initial starting value.
#9 Switch back to MDI mode select the offset menu then press key “Part Set Zero” Again
#10 Repeat the procedure for the remaining X,Y face

Zeroing Tools-
#1 Now we are ready to zero the tools look back at your SOLIDWORKS CAM file open the Setup Machine option and select the tool menu (This will indicate what tools are used within your program and need to be zeroed)
#2 Switch to MDI mode after moving the tool away from the part in the positive Z axis via hang jog. If the machine is not already in it enter desired tool number ex. T1 (T + 1 + ATC FWD)
#3 Return to Hand Jog mode
#4 Place a piece of printer paper on the top face of your part
#5 slowly lower the tool until the printer paper can barely slide under its tip It is recommended to complete this step in small increments to prevent a crash
#6 Return to MDI mode select the offset menu twice to move to the tool offset menu *You should be automatically selected to the correct tool
#7 Select the Z axis and hit the “Tool Offset Measure” key to zero the tool
#9 Retrurn to hand jog mode move tool in positive Z to clear part
#8 Repeat for all other listed program tools
#10 You are now ready to run your part

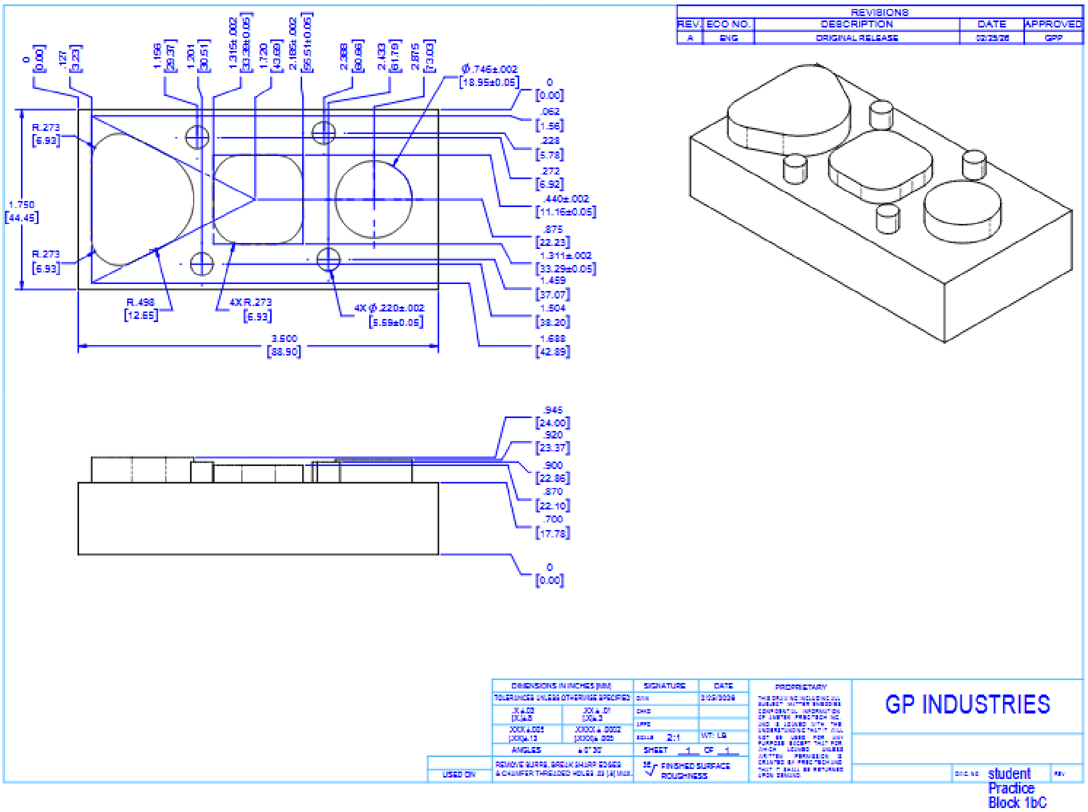
Running parts

#1 Export G-code from SOLIDCAM using the “Post Process” button (Make sure Haas VF3 is the selected post processor) in the .nc format place onto a USB storage device
#2 Select the List Program Button
#3 Use arrow keys to navigate to the USB mass storage device
#4 scroll down with arrow keys until you find the desired program
#5 Click “Select Program” and load the program into the machines internal memory
#7 Press “Mem”
#8 Press “Setting Graph” twice to verify the program and watch a visual preview
#9 Limit Rapid movement speed to 5% to prevent crashes
#10 Once program has been verified you are ready to run back out of the cycle start menu and close the door you may then press “Cycle Start” to begin with the Mem mode selected
#11 Watch the first pass to see if any adjustments must be made you may need to alter the position of the part within the stock in order to avoid crashes

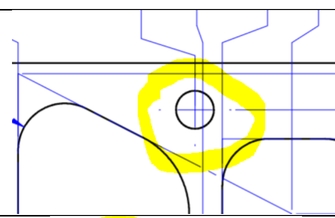
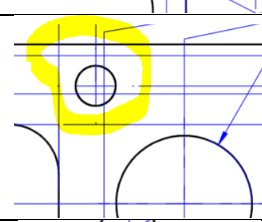
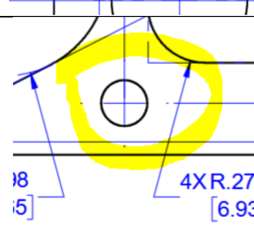
Inspection – Male Part

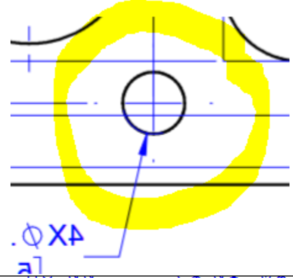
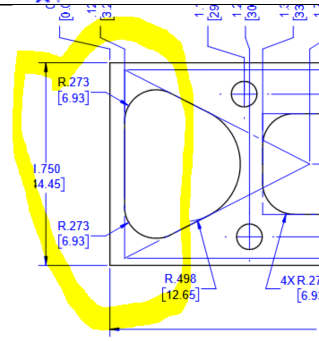
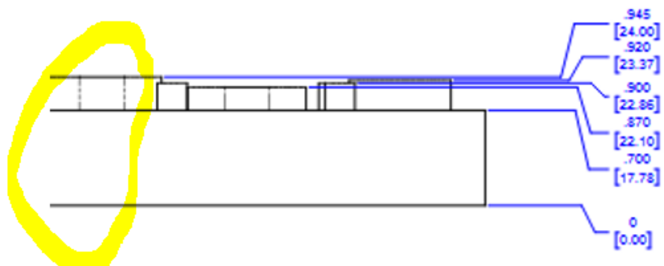
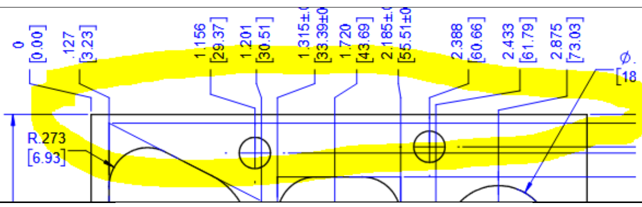
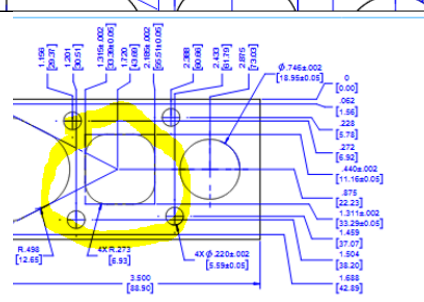
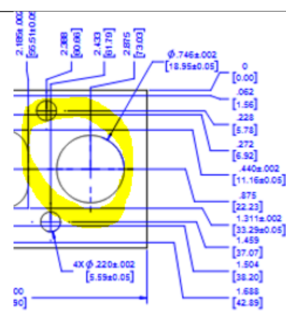
*Note All features on the Z face of part have been produced using a 0.19in endmill.

Print



Part Inspection

Feature	Measured (mm)	Photo / Print Zoomed View
Pin #1 (Top left)	Height:5.01mm (From Surface) 22.79mm (From Datum 0)	
Pin #2 (Top right)	Height:5.14mm (From Surface) 22.92mm (From Datum 0)	
Pin #3 (Bottom Left)	Height:5.12mm (From Surface) 22.90mm (From Datum 0)	

Pin #4 (Bottom right)	Height:5.14mm (From Surface) 22.92mm (From Datum 0)	
Width	44.45mm	
Height	24.09mm	
Length	88.85mm	
Central Rounded rectangle peg	Height:4.39mm (From Surface) 22.17mm (From Datum 0) Width:22.22mm Length 22.12mm	
Right Large Circular peg	Height:6.31mm (From Surface) 23.09mm (From Datum 0)	

Left rounded triangle peg	Height:6.21mm (From Surface) 23.99mm (From Datum 0)